

Paired 7/28-day compressive-strength model

30 mixtures × 2 curing ages · linear mixed model (random mixture intercept, age-specific residual variance) · corrected data set · generated 2026-09-28

Findings

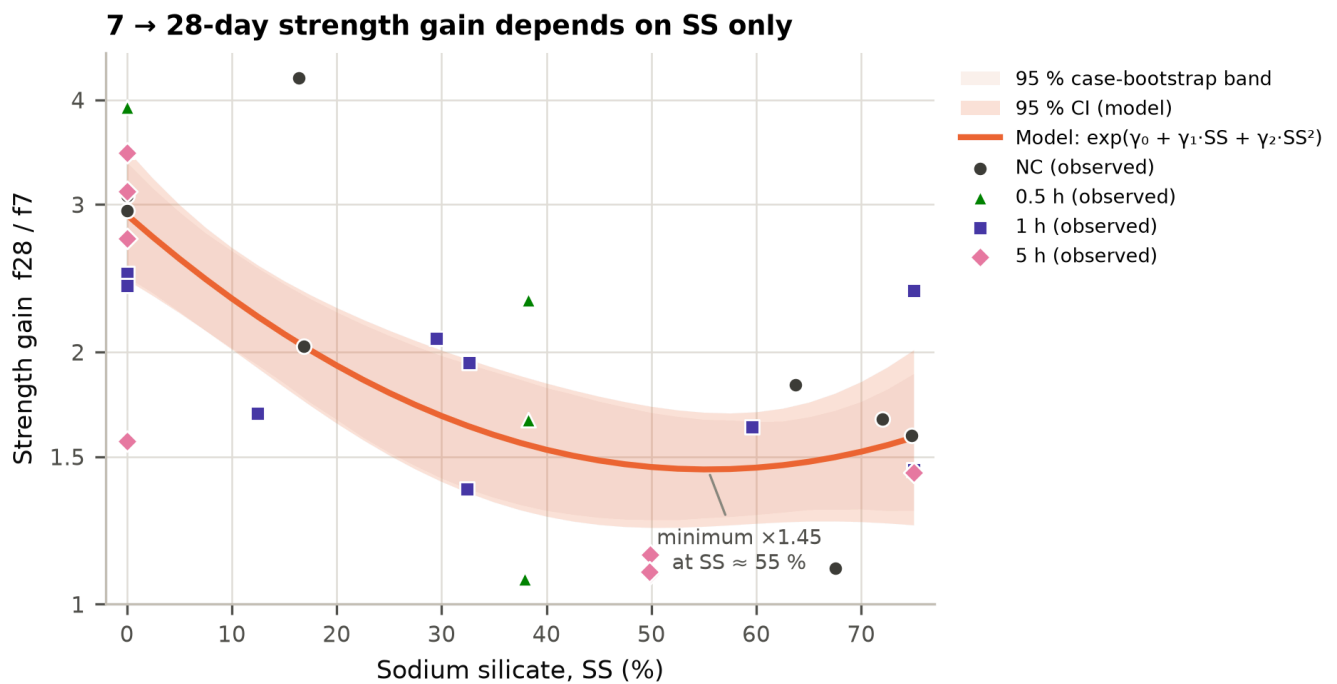
Robust · Strength rises with SS to a peak at mid-range SS, at both ages SS and SS ² are the largest terms (p < 0.001, p < 0.001) and are kept in at least 99 % of re-selections. The peak lies at SS 58 % (7 d) and 61 % (28 d) for uncarbonated mixes at RCF 30 %, and lower (SS 46 % at 7 d, 38 % at 28 d) for carbonated mixes at RCF 50 %.
Robust · The 7 → 28-day gain is set by SS $f_{28}/f_7 = 2.92$ (95 % CI 2.44–3.49) at SS 0 %, falls to a minimum of 1.45 near SS 55 % and is 1.58 (1.24–2.01) at SS 75 % (SS × Age p < 0.001). NaOH-only mixes gain most. SS is the only age-dependent variable in 86 % of residual-bootstrap and 91 % of subsample re-selections. The curvature of the gain (SS ² × Age, p = 0.017) is moderate.
Moderate · Carbonated RCF helps in NaOH-rich mixes and hurts in silicate-rich ones Carbonated × SS p < 0.001. At RCF 50 % and SS 0 % a 1 h carbonated mix is ×1.55 (95 % CI 1.18–2.05; p = 0.003) the uncarbonated one; at RCF 10 % and SS 75 % it is ×0.53 (95 % CI 0.40–0.72; p < 0.001). The crossover SS rises with RCF, from 5 % at RCF 10 % to 49 % at RCF 50 % (1 h). Kept in 97 % of residual-bootstrap but only 67 % of subsample re-selections: it rests on the 8 uncarbonated mixes.
Moderate · RCF raises strength when it is carbonated and when SS is low Carbonated × RCF p = 0.024, RCF × SS p = 0.026. Going from 10 to 50 % RCF multiplies strength by ×2.08 (95 % CI 1.64–2.64; p < 0.001) for carbonated RCF at SS 0 %, and by ×0.91 (95 % CI 0.65–1.26; p = 0.55) for uncarbonated RCF at SS 75 %. RCF never lowers strength significantly. The case-bootstrap intervals of carbonated × RCF and RCF × SS include zero, so these two terms depend on a few mixtures.
Not detected · Carbonation duration (0.5, 1, 5 h): no difference detected The three carbonated levels do not differ in level (p = 0.12); on/off coding fits about equally well (ΔAICc −1.13). Carbonation × age is not supported (p = 0.19). In the age-specific fit the 5 h level gains less than NC (×0.77, 95 % CI 0.58–1.02; p = 0.069), a single contrast that the joint test does not support.
Not detected · A/B ratio (0.42–0.48): no effect detected No A/B term improves the model at either age (A/B p = 0.68, (A/B) ² p = 0.29, A/B × Age p = 0.31). A/B curvature is selected only if the 28-day variance is forced equal to the 7-day variance, and does not survive nested cross-validation.

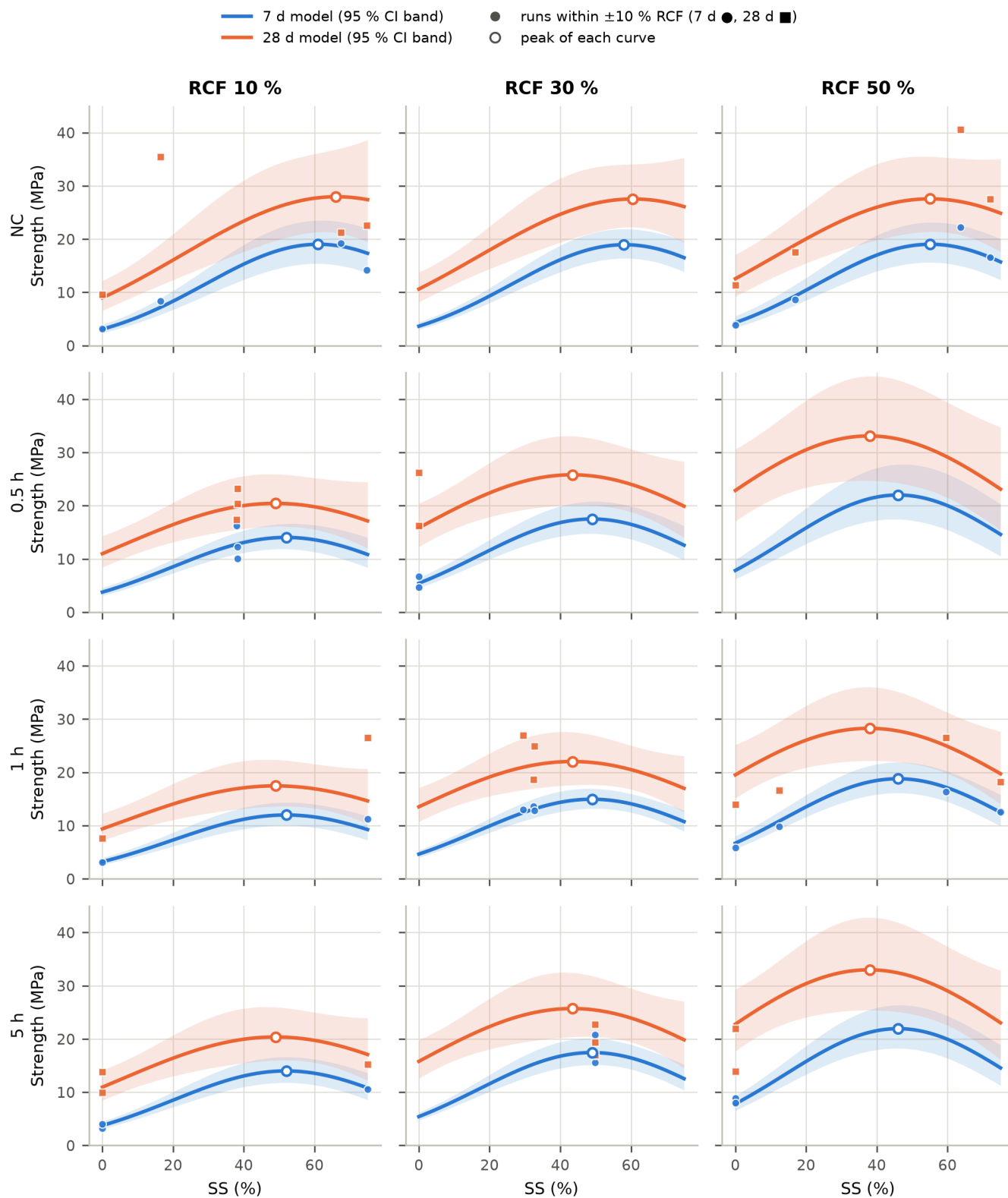
Final model

$\ln f = 2.9551 + 0.0598 \cdot A + 0.6029 \cdot D + 0.0650 \cdot C_{0.5h} - 0.0931 \cdot C_{1h} + 0.0613 \cdot C_{5h} - 0.1085 \cdot AD - 0.5249 \cdot D^2 + 0.1982 \cdot K \cdot A - 0.3360 \cdot K \cdot D + g \cdot (0.4435 - 0.3059 \cdot D + 0.3208 \cdot D^2)$			
$\ln(f_7) = 1.03971 + 0.0084169 \cdot RCF + 0.0610455 \cdot SS - 1.4469 \times 10^{-4} \cdot RCF \cdot SS - 4.8732 \times 10^{-4} \cdot SS^2 + 0.103748 \cdot C[0.5 \text{ h}] - 0.054418 \cdot C[1 \text{ h}] + 0.100028 \cdot C[5 \text{ h}] + 0.00991027 \cdot K \cdot RCF - 0.00896104 \cdot K \cdot SS$			
$\ln(f_{28}) = 2.10988 + 0.0084169 \cdot RCF + 0.0357793 \cdot SS - 1.4469 \times 10^{-4} \cdot RCF \cdot SS - 2.5922 \times 10^{-4} \cdot SS^2 + 0.103748 \cdot C[0.5 \text{ h}] - 0.054418 \cdot C[1 \text{ h}] + 0.100028 \cdot C[5 \text{ h}] + 0.00991027 \cdot K \cdot RCF - 0.00896104 \cdot K \cdot SS$			
$\ln(f_{28}/f_7) = 1.07017 - 0.0252662 \cdot SS + 2.2811 \times 10^{-4} \cdot SS^2$			
7 d · R ² 0.95 pred. R ² 0.90 · RMSE 2.2 MPa	28 d · R ² 0.34 pred. R ² 0.09 · RMSE 7.7 MPa	Residual scatter 15 % / 34 % 7 d / 28 d (replicate 19 % / 20 %)	7 ↔ 28 d correlation 0.49 within mixture (paired)

Term	df	F	p	Residual bootstrap	Subsamples 24/30	Evidence
RCF	1, 20.0	0.83	0.37	100 %	99 %	Retained for hierarchy
SS	1, 26.3	65.78	< 0.001	100 %	100 %	Robust
Carbonation (4 levels)	3, 20.0	1.59	0.22	98 %	71 %	Retained for hierarchy
RCF × SS	1, 20.0	5.83	0.026	82 %	47 %	Moderate
SS ²	1, 21.4	28.66	< 0.001	100 %	100 %	Robust
Carbonated × RCF	1, 20.0	5.96	0.024	81 %	41 %	Moderate
Carbonated × SS	1, 20.0	16.85	< 0.001	97 %	67 %	Moderate
Age (28 vs 7 d)	1, 27.0	24.38	< 0.001	–	–	Design factor
SS × Age	1, 27.0	18.83	< 0.001	100 %	100 %	Robust
SS ² × Age	1, 27.0	6.44	0.017	75 %	62 %	Moderate

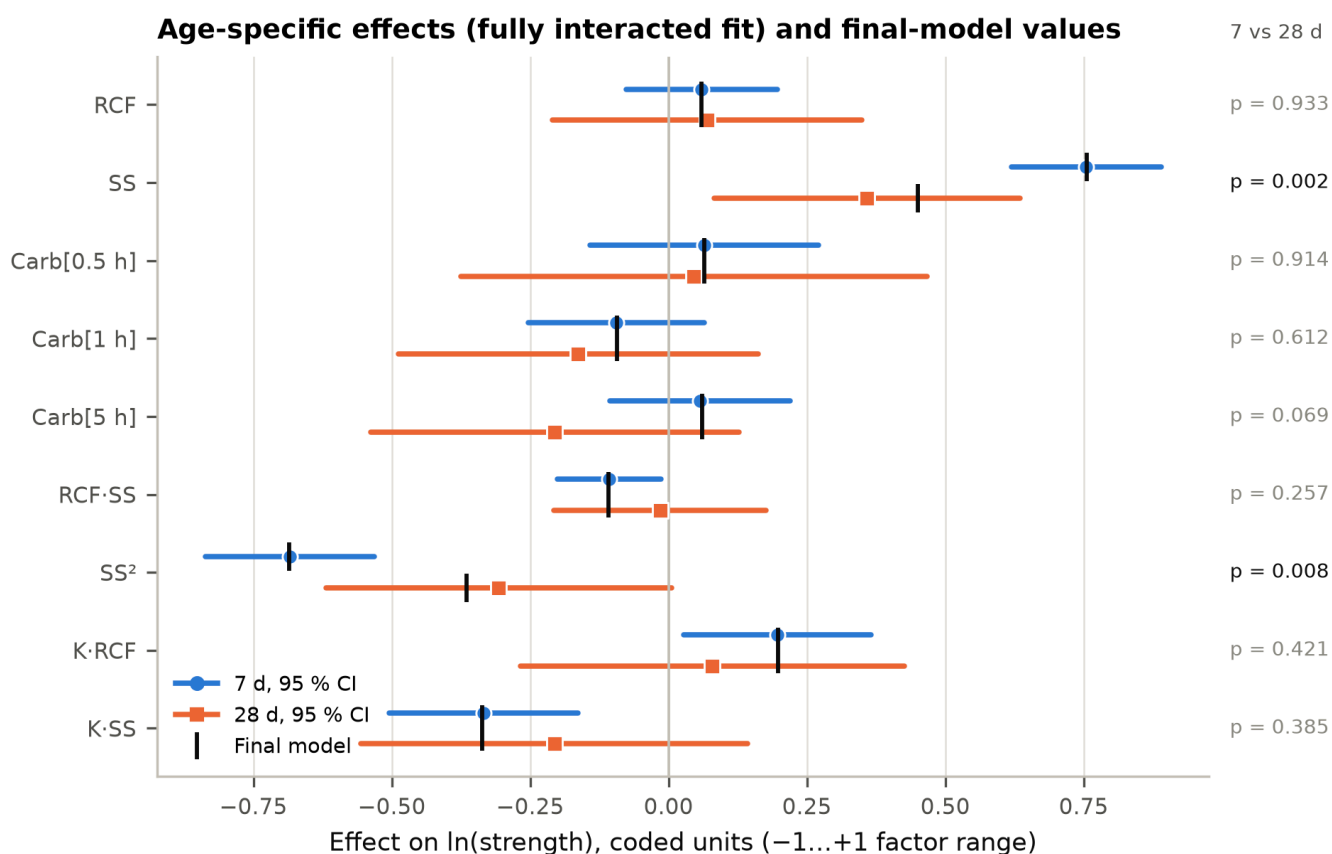
Wald F tests, Satterthwaite df. Covariance UN: SD 0.153 (7 d), 0.327 (28 d), $\rho = 0.49$. Lack of fit: gain $p = 0.35$, level/gain $p = 0.65$. Box-Cox (final): $\lambda = 0.13$ (95 % CI -0.11 to 0.40). Exhaustive search of 53,105 models; ΔAICc to best CS model 4.9.





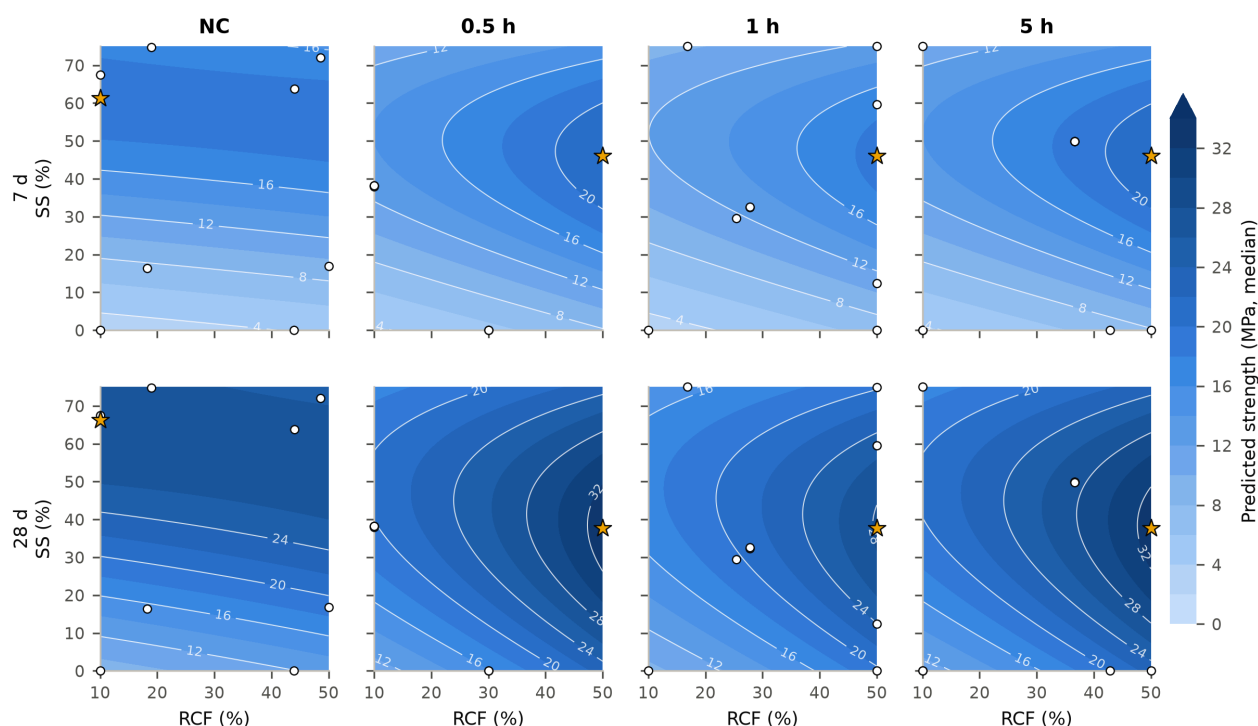
Age-specific effects

Term (coded)	Effect at 7 d [95 % CI]	Effect at 28 d [95 % CI]	Change 7→28 d	p (change)	Age term in final model
RCF	0.060 [−0.077, 0.197]	0.070 [−0.210, 0.349]	0.010 [−0.230, 0.250]	0.93	no (common effect)
SS	0.754 [0.619, 0.890]	0.358 [0.082, 0.635]	−0.396 [−0.633, −0.159]	0.002	yes
Carb 0.5 h	0.065 [−0.141, 0.271]	0.046 [−0.375, 0.467]	−0.019 [−0.381, 0.343]	0.91	no (common effect)
Carb 1 h	−0.094 [−0.253, 0.065]	−0.163 [−0.488, 0.162]	−0.069 [−0.348, 0.210]	0.61	no (common effect)
Carb 5 h	0.057 [−0.105, 0.220]	−0.205 [−0.537, 0.127]	−0.262 [−0.548, 0.023]	0.069	no (common effect)
RCF·SS	−0.107 [−0.201, −0.013]	−0.015 [−0.207, 0.177]	0.092 [−0.073, 0.257]	0.26	no (common effect)
SS ²	−0.684 [−0.837, −0.532]	−0.307 [−0.619, 0.005]	0.378 [0.109, 0.646]	0.008	yes
K·RCF	0.196 [0.027, 0.366]	0.079 [−0.267, 0.426]	−0.117 [−0.414, 0.180]	0.42	no (common effect)
K·SS	−0.334 [−0.505, −0.163]	−0.206 [−0.556, 0.143]	0.128 [−0.172, 0.427]	0.39	no (common effect)



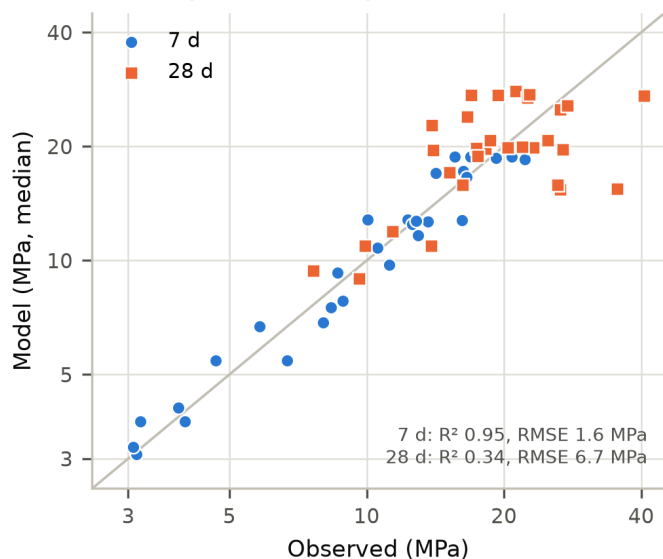
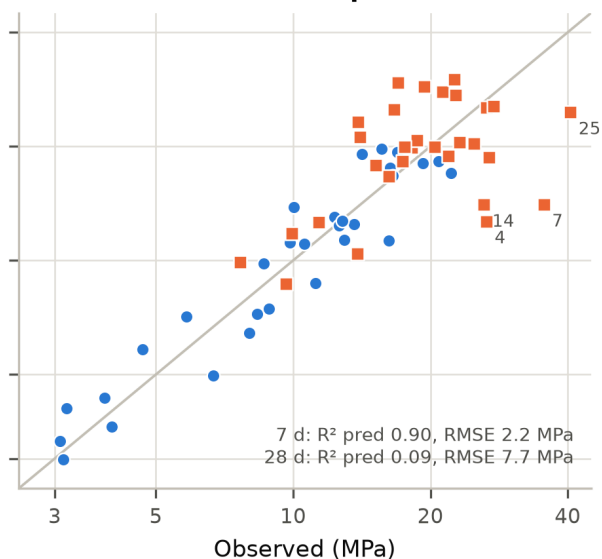
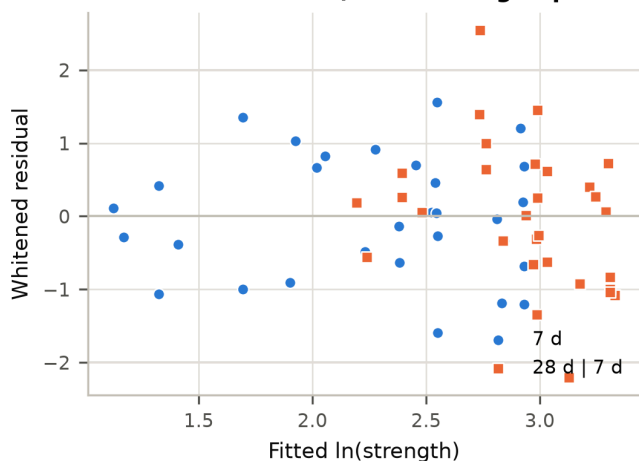
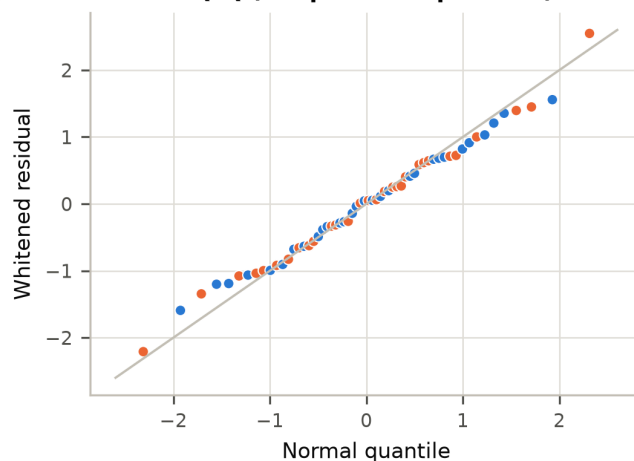
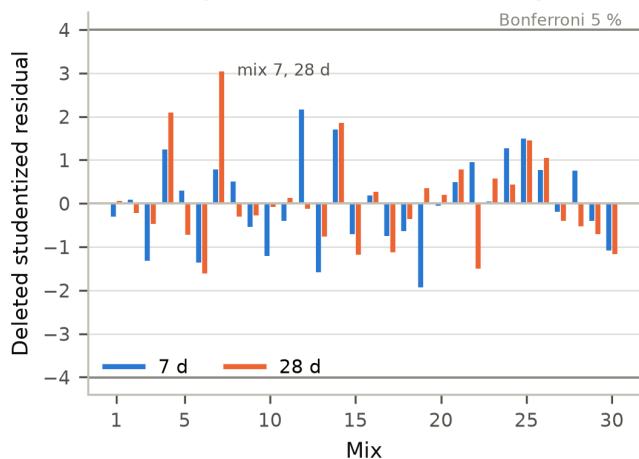
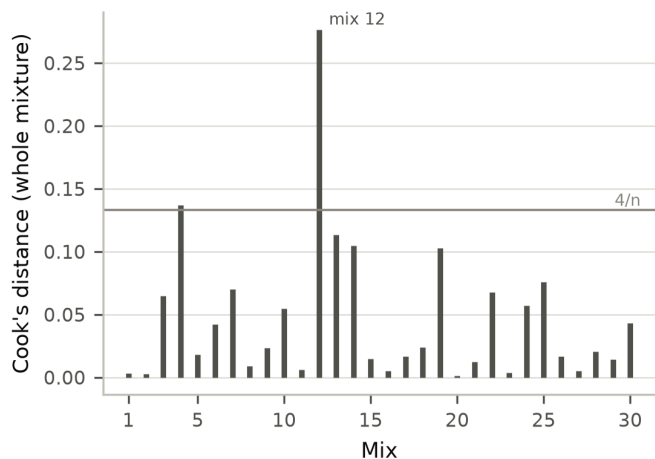
Age	Carbonation	Maximum at	Median (MPa)	95 % CI	95 % PI (one batch)	Within 5 % of maximum	Scaled pred. variance / design max
7 d	NC	RCF 10 %, SS 61.2 %	19.0	15.4–23.6	13.0–28.0	RCF 10–50, SS 45–71	0.45 / 0.60
7 d	0.5 h	RCF 50 %, SS 46.0 %	22.0	17.4–27.8	14.8–32.7	RCF 46–50, SS 36–56	0.54 / 0.60
7 d	1 h	RCF 50 %, SS 46.0 %	18.8	16.1–21.9	13.2–26.8	RCF 46–50, SS 36–56	0.24 / 0.60
7 d	5 h	RCF 50 %, SS 46.0 %	21.9	18.2–26.4	15.2–31.7	RCF 46–50, SS 36–56	0.34 / 0.60
28 d	NC	RCF 10 %, SS 66.2 %	28.0	21.3–36.7	13.6–57.3	RCF 10–50, SS 43–75	0.17 / 0.25
28 d	0.5 h	RCF 50 %, SS 37.8 %	33.1	24.8–44.1	16.1–68.3	RCF 46–50, SS 24–52	0.19 / 0.25
28 d	1 h	RCF 50 %, SS 37.8 %	28.3	22.2–35.9	13.9–57.4	RCF 46–50, SS 24–52	0.13 / 0.25
28 d	5 h	RCF 50 %, SS 37.8 %	33.0	25.5–42.7	16.2–67.3	RCF 46–50, SS 24–52	0.15 / 0.25

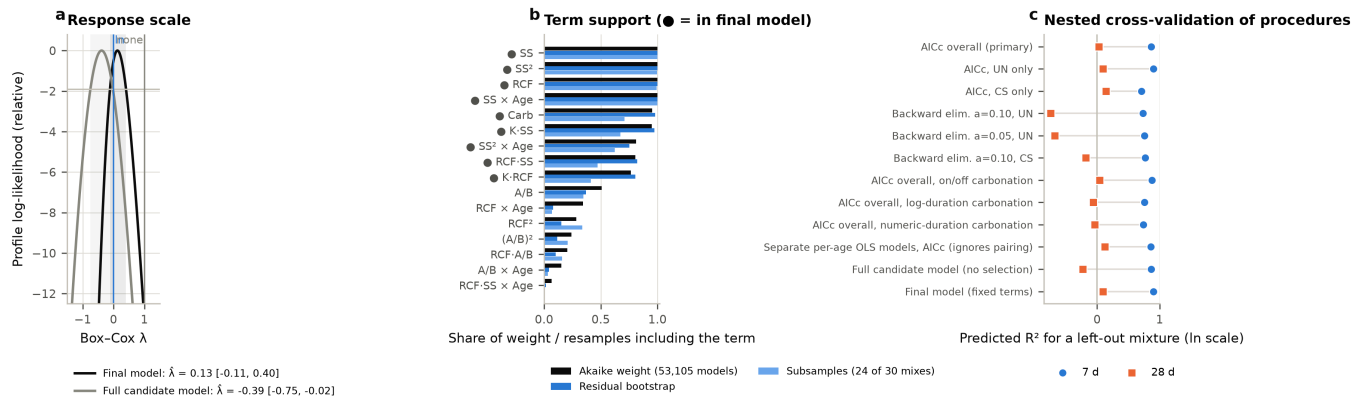
Predicted strength over RCF × SS (A/B not in model). ○ runs at that carbonation level, ★ maximum



Validation and robustness

Leave-one-mixture-out predicted R^2 : 0.90 (7 d), 0.09 (28 d); RMSE 2.2 / 7.7 MPa. Nested CV of the selection procedure: 0.87 / 0.03. Residual-bootstrap re-selection keeps SS as the only age-dependent variable in 86 % of resamples. Shapiro-Wilk $p = 0.95$; Breusch-Pagan $p = 0.64$; largest $|t|$ 3.05 (mix 7, 28 d; Bonferroni $p = 0.42$).

a Fitted (all 30 mixes)**b Leave-one-mixture-out prediction****a Residuals vs fitted (Breusch-Pagan $p = 0.64$)****b Normal Q-Q (Shapiro-Wilk $p = 0.95$)****c Outliers (mixture left out of the fit)****d Influence of each mixture on the coefficients**



Robustness: key model quantities under alternative analyses (black = primary)

